

TrES-3b

R-0466

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180801 · created 2026-07-20T00:03:12+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458331.7488 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1 +/- 0.054
Transit depth (Rp/Rs)^2	0.99 +/- 1.07 [%]
Orbital Inclination (inc)	79.93 +/- 2.93
Airmass coefficient 1 (a1)	1.024 +/- 0.06
Airmass coefficient 2 (a2)	-0.03 +/- 0.14
Scatter in the residuals of the lightcurve fit is	5.82 %
Best Comparison Star	#1 - [404, 397]
Optimal Aperture	3.87
Optimal Annulus	13.69
Transit Duration (day)	0.024 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1 ± 0.054 vs 0.16309 (deviation 0.97σ)
Duration fit vs published	0.024 d vs 0.0567 d (deviation 1.0σ)
Mid-time O–C	2.7 min
Depth SNR	1.5
Residual scatter	5.82 %

Light curve

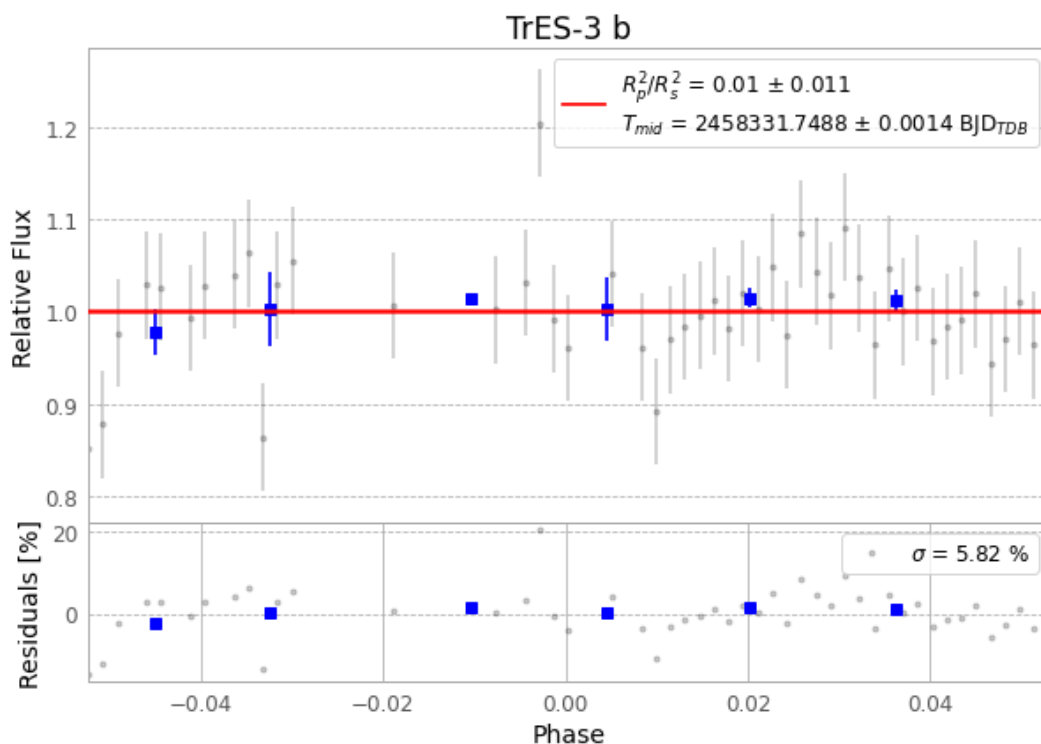


Figure 1 — FinalLightCurve_TrES-3 b_2018-07-31.png (SHA-256 7ecdb59f6f3ee400...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 df6f639aa50b539d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

67 FITS frames (44 MB), TRES-3180801041512.FITS → TRES-3180801073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180801.log (fafefcbab45c...), FinalLightCurve_TrES-3 b_2018-07-31.png (7ecdb59f6f3e...), FinalLightCurve_TrES-3 b_2018-07-31.csv (a12fd34fd74e...)

Compute resources

Wall time 215.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 00:03Z.